Hemodynamic Disorders
SGL 6
Hyperemia, congestion and thrombosis

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Outline:

- Hyperemia and congestion
- Thrombosis
  - ✓ Virchow's triad
  - ✓ Morphology of thrombus
  - ✓ Fate of thrombus
  - ✓ Clinical significance
- Summary

Hyperemia and Congestion

Hyperemia and congestion both refer to an increase in blood volume within a tissue, but have different underlying mechanisms

Hyperemia

- is an active process in which **arteriolar dilation** leads to **increased blood flow.**
- Affected tissues turn red (erythema) because of the engorgement of vessels with **oxygenated blood.**

Example of hyperemia:

- sites of inflammation



- in skeletal muscle during exercise



Congestion

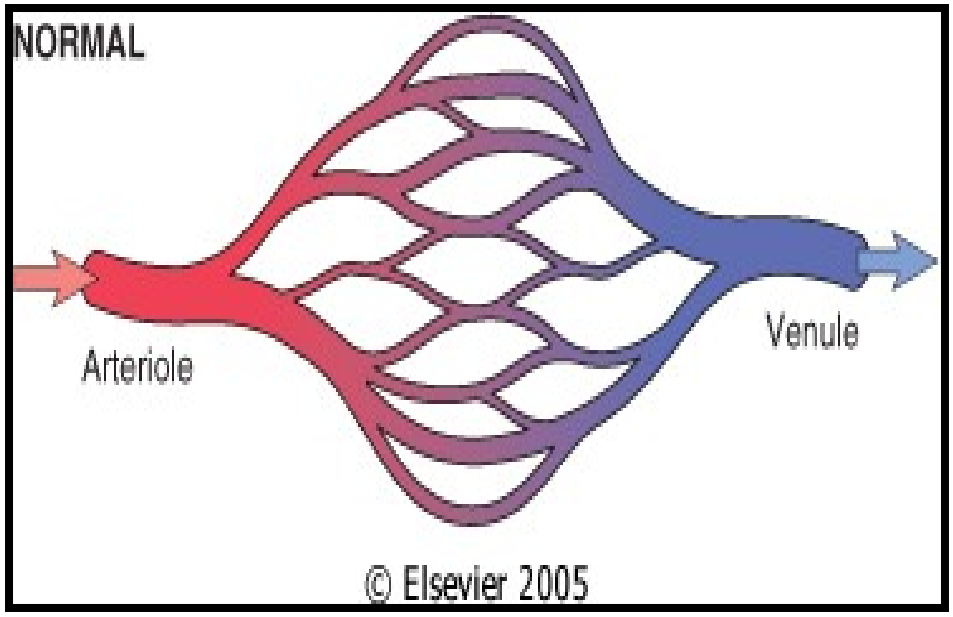
- is a passive process
- resulting from **impaired outflow** of **venous blood** from a tissue.
- Congested tissues have an abnormal blue-red color (cyanosis) that stems from the accumulation of **deoxygenated hemoglobin** in the affected area.

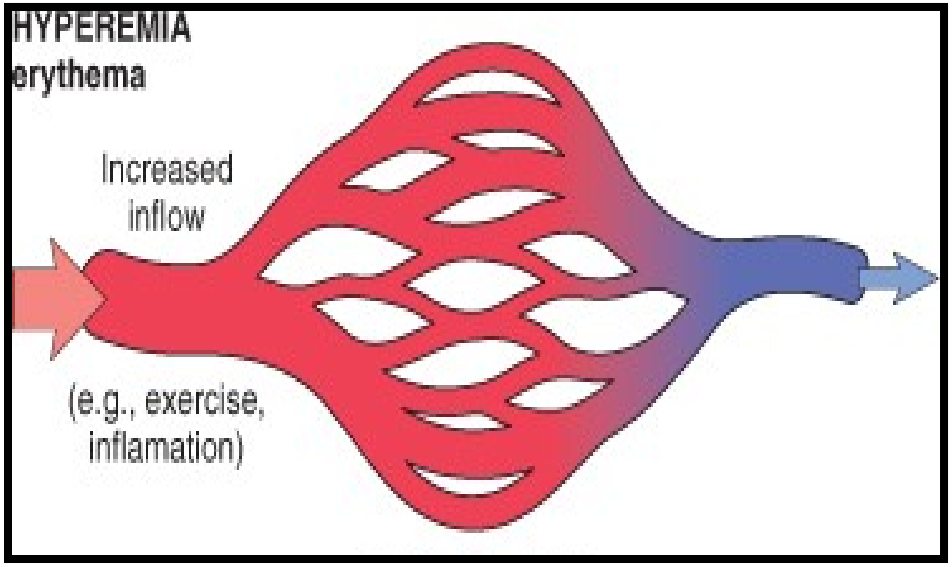


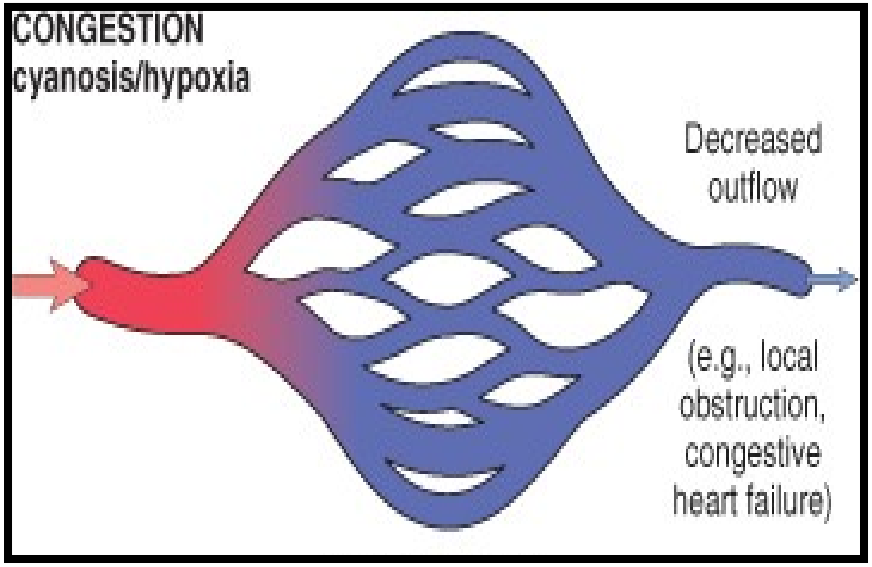
Example of congestion:

- It can occur systemically, as in cardiac fa...ure
- or locally as a consequence of an isolated venous obstruction.




NORMAL
Arteriole
Venule
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HYPEREMIA
erythema
Increased
inflow
(e.g., exercise,
inflamation)


CONGESTION
cyanosis/hypoxia
Decreased
outflow
(e.g., local
obstruction,
congestive
heart failure)

Thrombosis

It is the **pathologic counterpart** of hemostasis, define as the formation of blood clot (thrombus) within **non-traumatized, intact vessels.**

((Remember that **normal hemostasis** comprises a series of regulated processes that culminate in the formation of a blood clot that limits bleeding from an injured vessel))

Hemostasis is a precisely orchestrated process involving **platelets, clotting factors, and endothelium** that occurs at the site of **vascular injury** and culminates in the formation of a blood clot, **which serves to prevent or limit the extent of bleeding.**

The primary abnormalities that lead to intravascular thrombosis are:

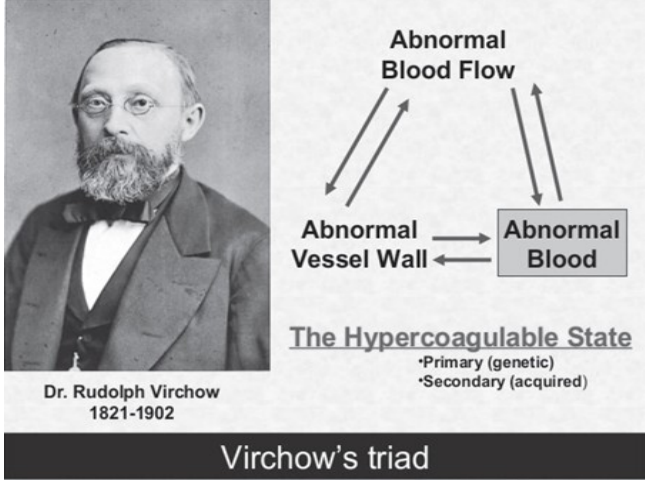
(1) endothelial injury,

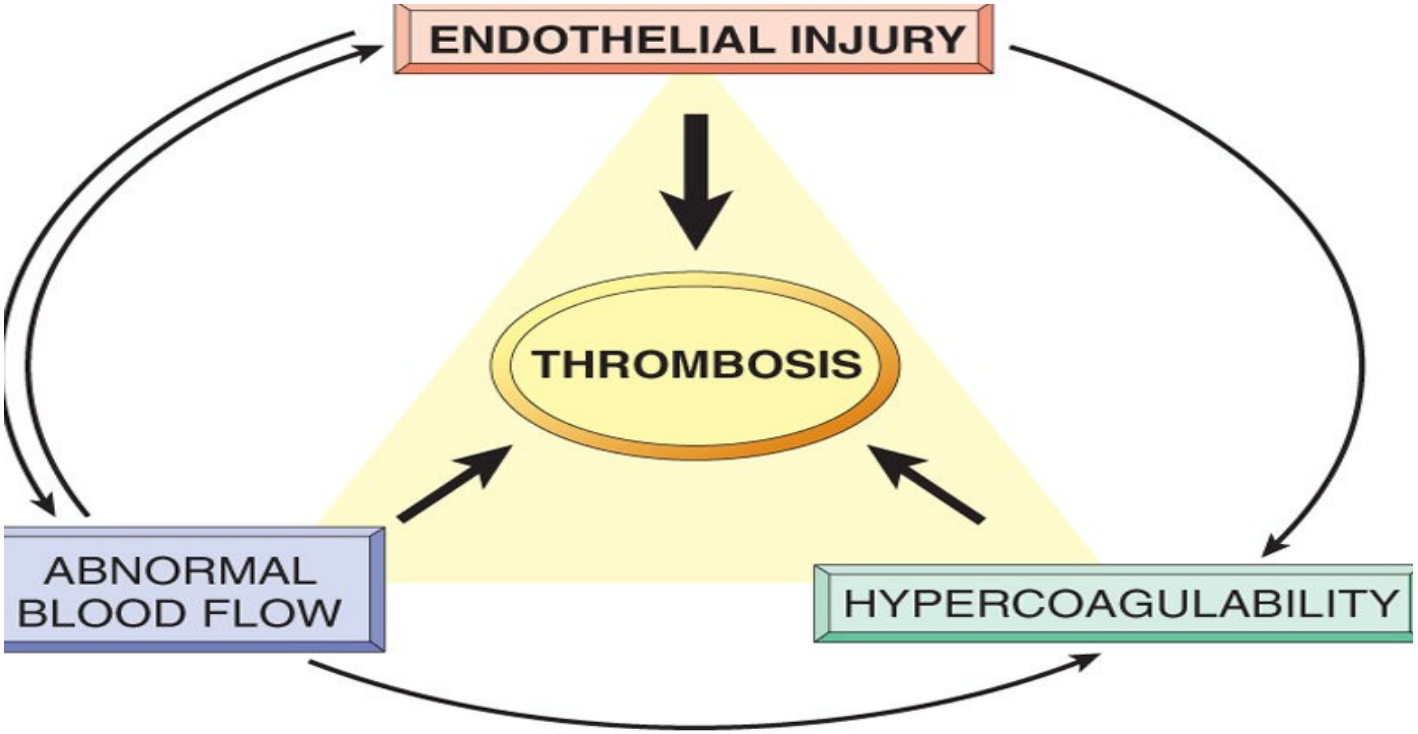
(2) stasis or turbulent blood flow, and

(3) hypercoagulability of the blood

(the so-called “Virchow triad”)

Virchow's triad:





1- **Endothelial injury**

endothelial injury or **dysfunction** can be produced by diverse exposures, including:

1.physical injury,

2. infectious agents

3.abnormal blood flow

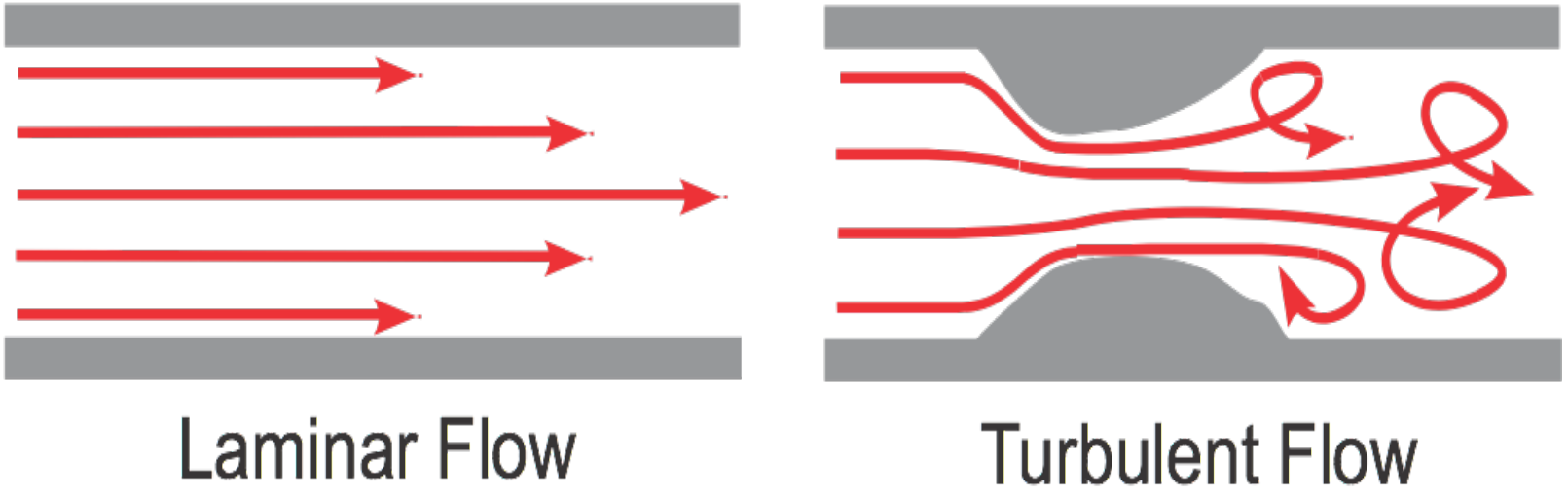
4.inflammatory mediators

5.metabolic abnormalities, such as hypercholesterolemia

6.toxins absorbed from cigarette smoke.

2- Abnormal Blood Flow:

Turbulence contributes to arterial and cardiac thrombosis by causing endothelial injury or dysfunction, as well as by forming countercurrents and local pockets of **stasis.**



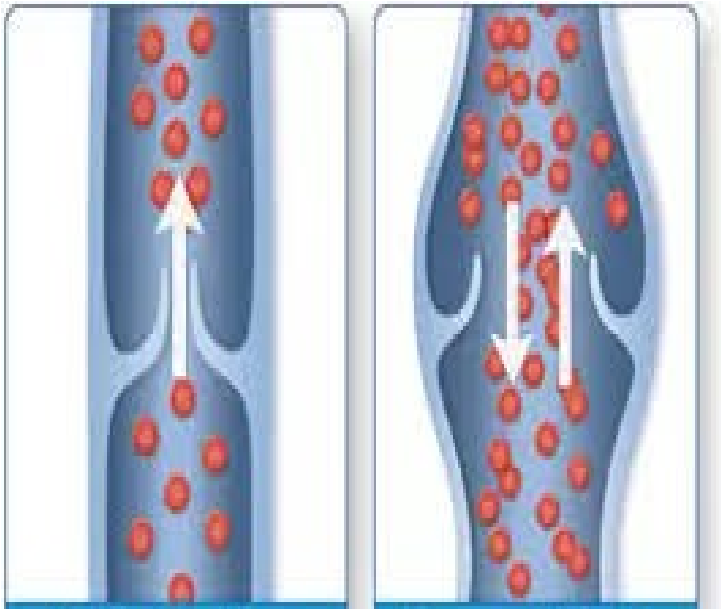
Stasis and turbulence effect:

a) Both promote endothelial cell activation and enhanced procoagulant activity, in part through flow-induced changes in endothelial gene expression.

b) Stasis allows platelets and leukocytes to come **into contact** with the endothelium when the flow is sluggish.

Note: (Under conditions of normal laminar blood flow, platelets and other blood cells are found mainly in the center of the vessel lumen, separated from the endothelium by a slower-moving layer of plasma)

c) Stasis also slows the washout of activated clotting factors and impedes the inflow of clotting factor inhibitors.



3. Hypercoagulable States

- Hypercoagulability refers to **an abnormally high tendency of the blood to clot**
- is typically caused by alterations in coagulation factors.
- It contributes infrequently to arterial or intracardiac thrombosis but is an important underlying risk factor for venous thrombosis.

WHY??

The alterations of the coagulation pathways that predispose affected persons to thrombosis can be divided into primary (genetic) and secondary (acquired) disorders:

Primary (Genetic):

Common (>1% of the Population):

1. Factor V mutation (G1691A mutation; factor V Leiden)
2. Prothrombin mutation (G20210A variant)
3. Increased levels of factor VIII, IX, or XI or fibrinogen

Rare:

1.Anti-thrombin III deficiency
2.Protein C deficiency
3. Protein S deficiency

Very Rare:

1.Fibrinolysis defects
2. Homozygous homocystinuria (deficiency of cystathione β-synthetase)

Secondary (Acquired):

<u>High Risk for Thrombosis:</u>

1. Prolonged bed rest or immobilization
2. Myocardial infarction
3. Atrial fibrillation
4. Tissue injury (surgery, fracture, burn)
5. Cancer
6. Prosthetic cardiac valves
7. Disseminated intravascular coagulation
8. Heparin-induced thrombocytopenia
9. Anti-phospholipid antibody syndrome

Lower Risk for Thrombosis:

1.Cardiomyopathy

2. Nephrotic syndrome

3. Hyperestrogenic states (pregnancy and postpartum)

4. Oral contraceptive use

5.Sickle cell anemia

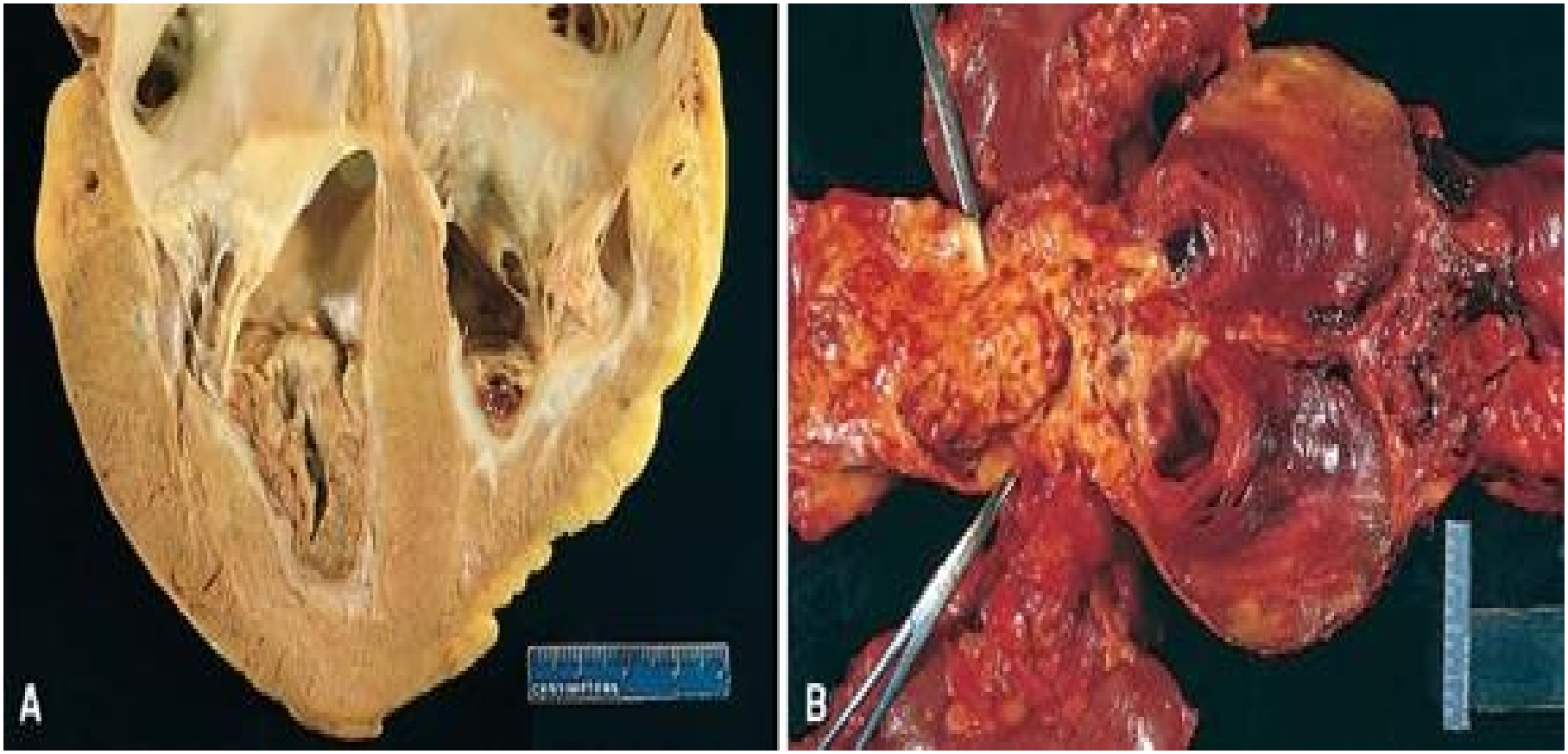
6. Smoking

Morphology of thrombus:

- Thrombi can develop **anywhere** in the cardiovascular system.
- Thrombi are **focally attached** to the underlying vascular surface
- Thrombi can have grossly (and microscopically) apparent laminations called **lines of Zahn:**

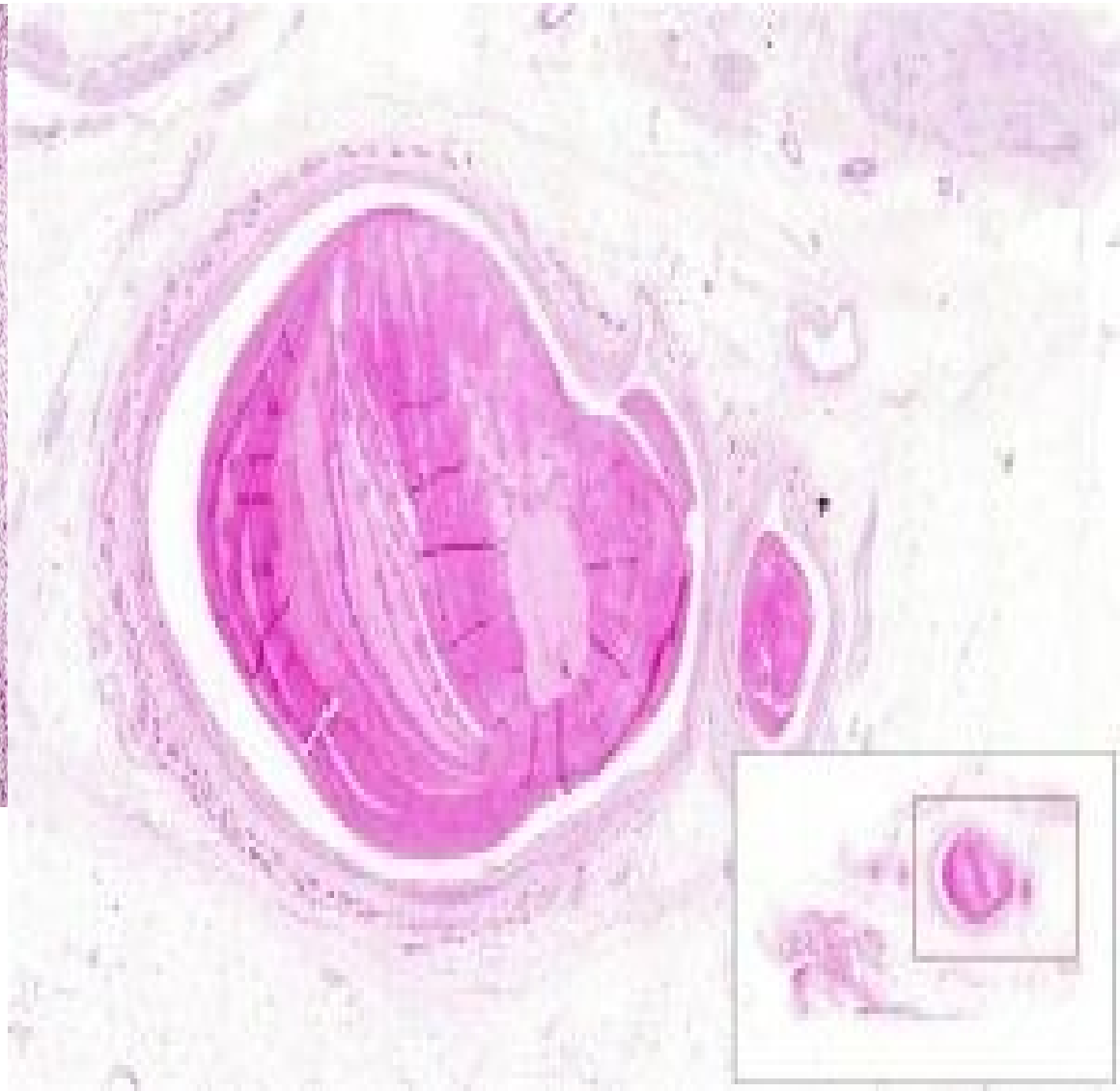
- these represent pale platelet and fibrin layers alternating with darker red cell–rich layers.
- Such lines are **significant** in that they are only found in thrombi that form in flowing blood; their presence can therefore usually distinguish antemortem thrombosis from the bland nonlaminated clots that form in the postmortem state.

- Thrombi occurring in heart chambers or in the aortic lumen are designated as **mural thrombi.**
- Thrombi on heart valves are called **vegetations.**



Mural thrombi. (A) Thrombus in the left and right ventricular apices, overlying white fibrous scar. (B) Laminated thrombus in a dilated abdominal aortic aneurysm.



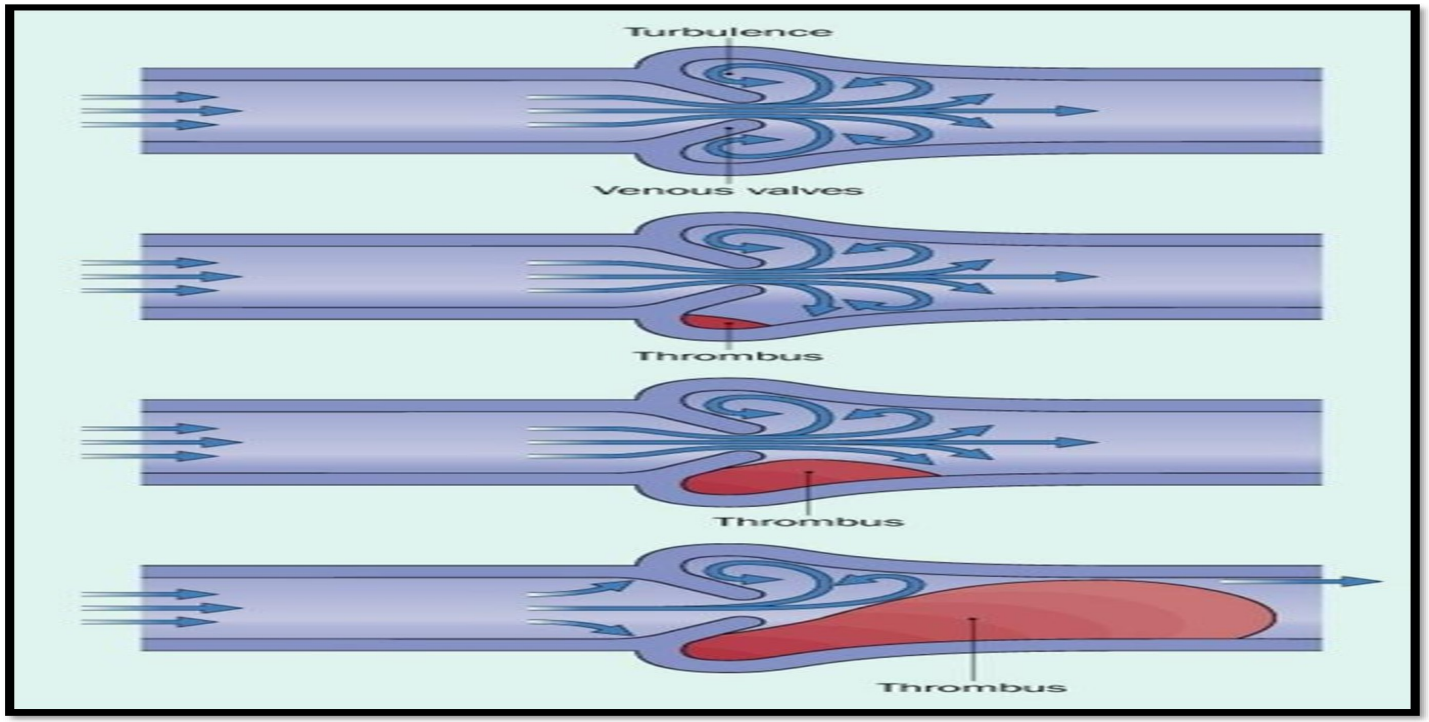


Lines of Zahn

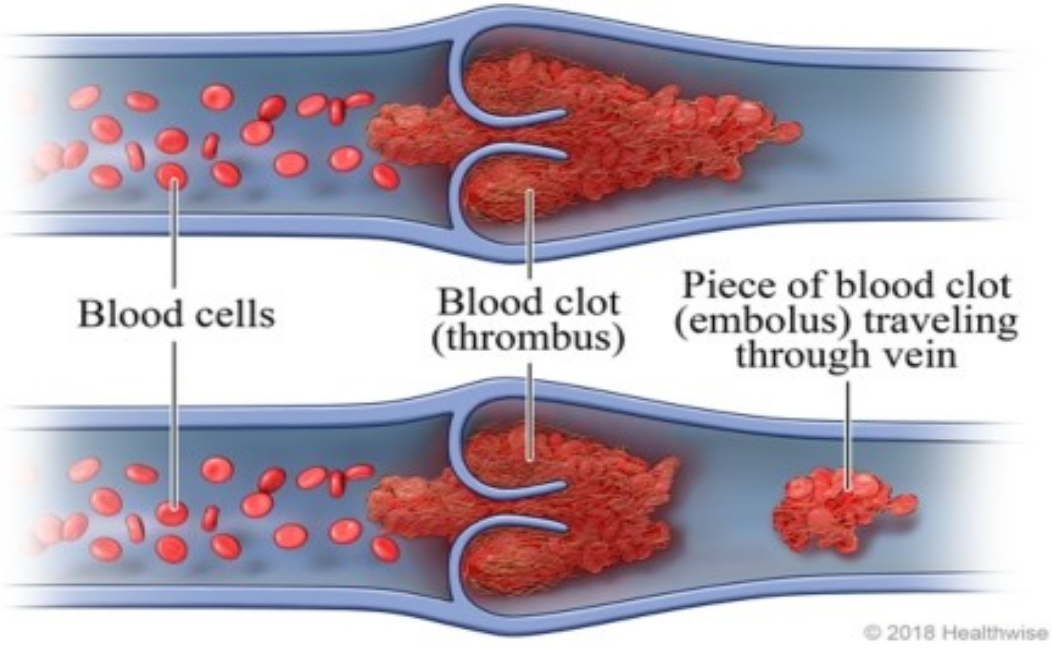
Fate of the Thrombus:

1)Propagation
2)Embolization
3)Dissolution
4)Organization and recanalization

1)Propagation: The thrombus enlarges through the accretion of additional platelets and fibrin, increasing the odds of vascular occlusion or embolization.

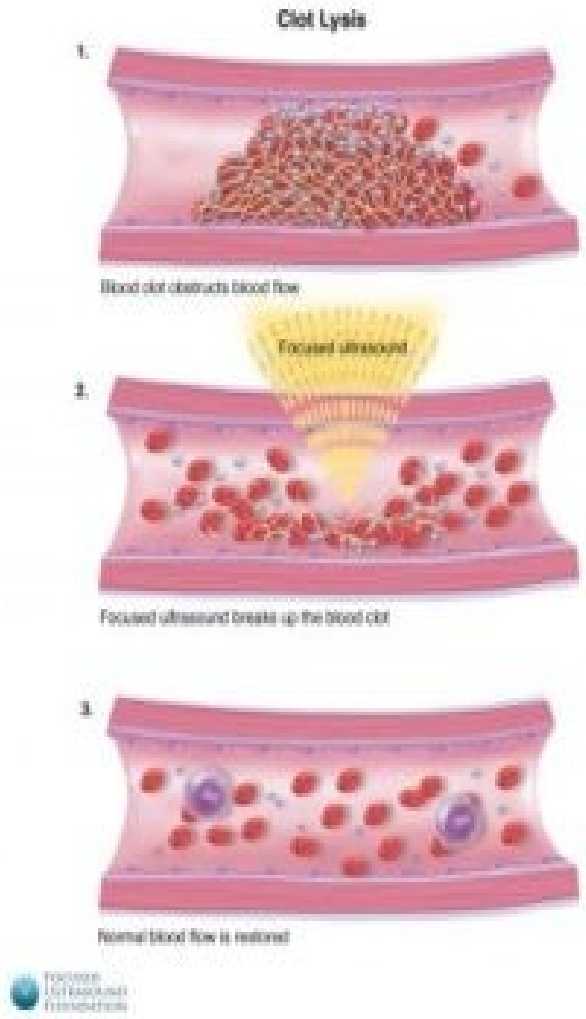


2)Embolization: Part or all of the thrombus is dislodged and transported elsewhere in the vasculature.



3)Dissolution:

✓If a thrombus is newly formed, activation of fibrinolytic factors may lead to its rapid shrinkage and complete dissolution.



✓With older thrombi, extensive fibrin polymerization renders the thrombus substantially more resistant to plasmin-induced proteolysis, and lysis is ineffectual.

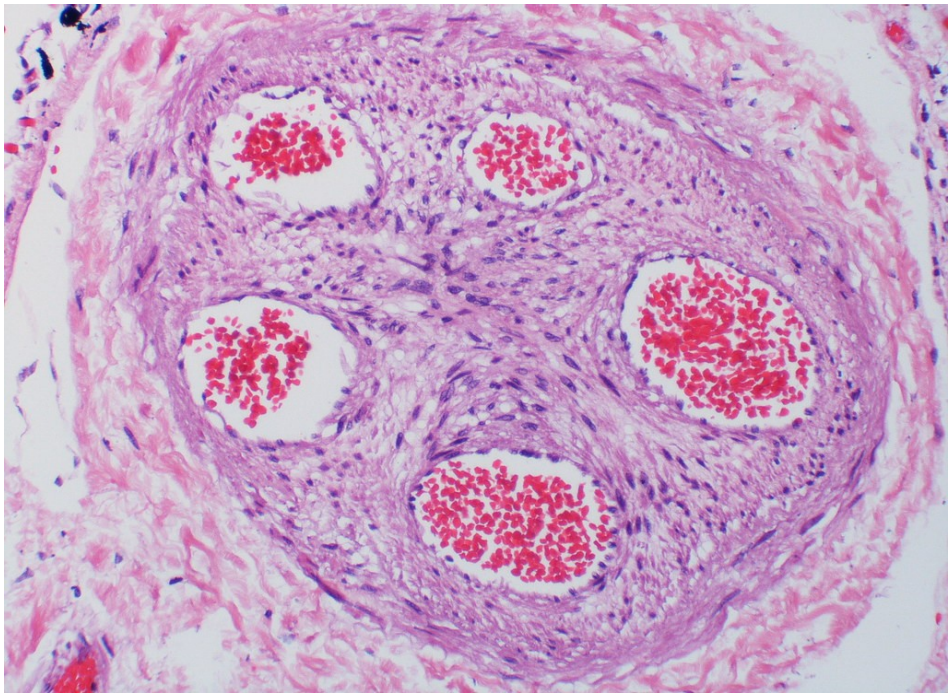
Clinical significance:

therapeutic administration of fibrinolytic agents generally is not effective unless administered within a few hours of thrombus formation.

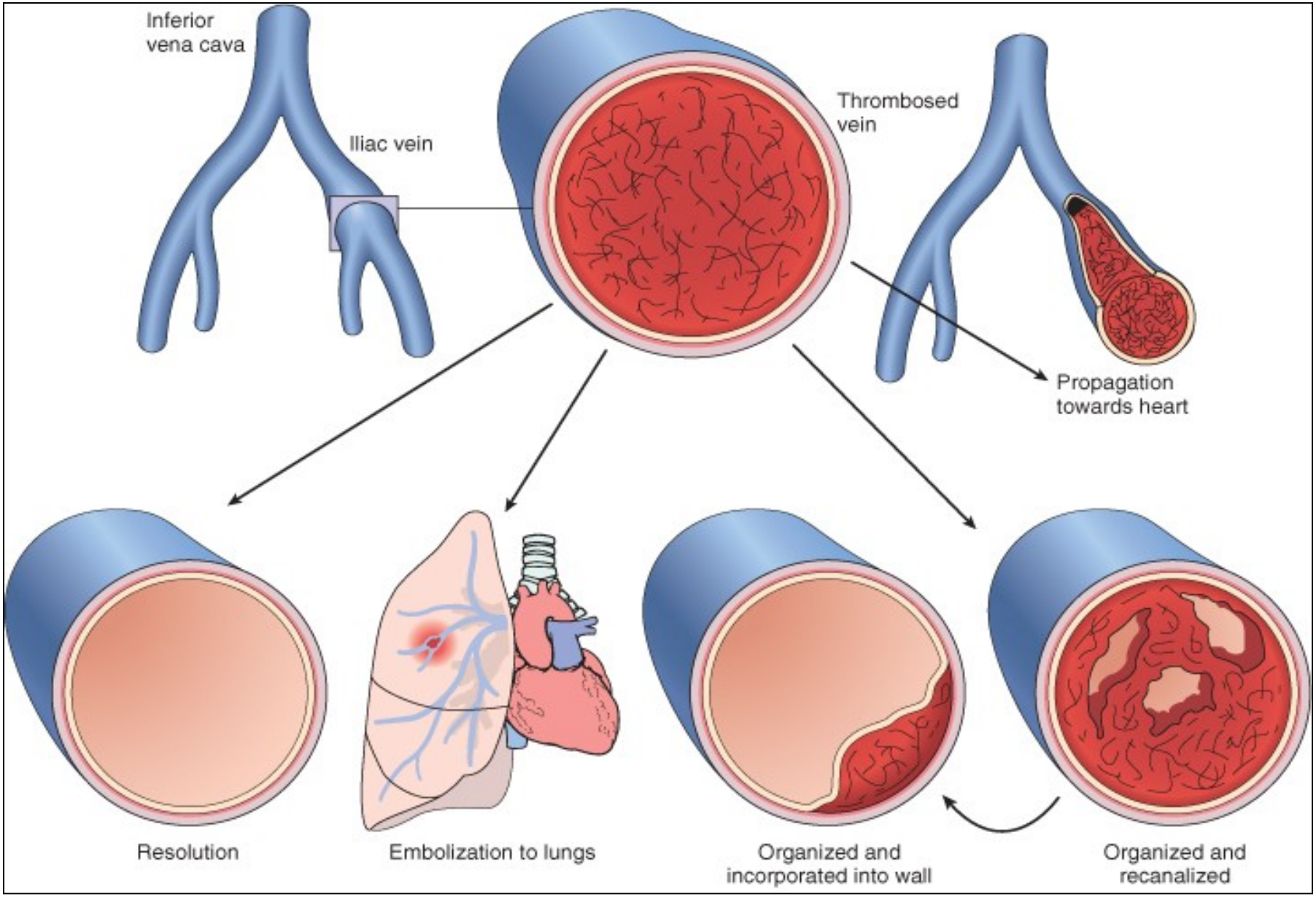
4)Organization and recanalization:

✓Older thrombi become organized by the ingrowth of endothelial cells, smooth muscle cells, and fibroblasts.

✓In time, capillary channels are formed that—to a limited extent—create conduits along the length of the thrombus, thereby reestablishing the continuity of the original lumen.



Fate of the Thrombus:



Clinical features:

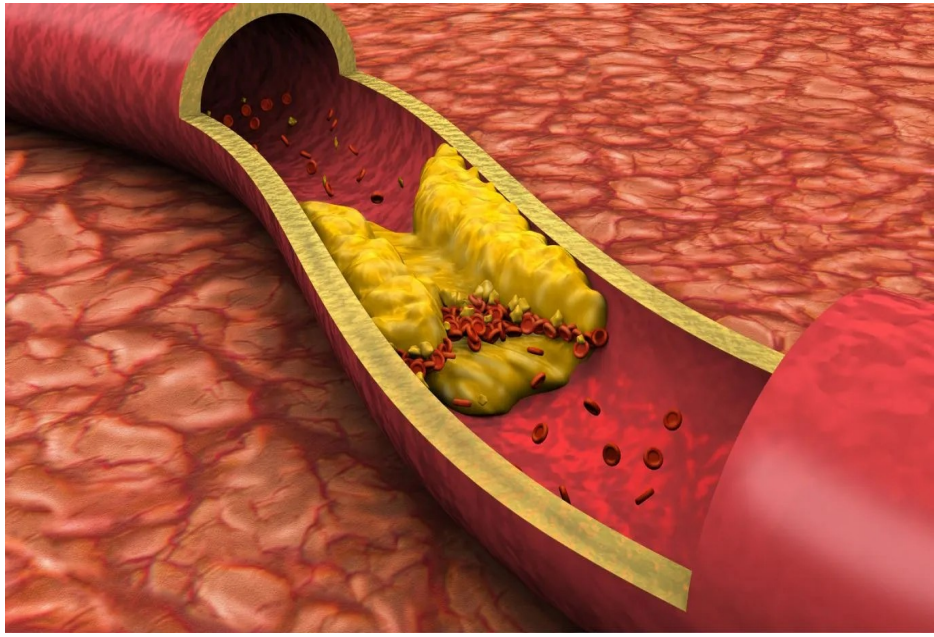
- The effect of thrombus depends on the site:

although venous thrombi can cause congestion and edema in vascular beds distal to an obstruction, they are most worrisome because of their potential to **embolize to the lungs and cause death.**

- Most venous thrombi occur in the superficial or the **deep veins of the leg.**



- Conversely, whereas arterial thrombi can embolize and cause tissue infarction, their **tendency to obstruct vessels** (e.g., in coronary and cerebral vessels) is considerably more important
- **Atherosclerosis** is a major cause of arterial thromboses



Summary:

- Thrombus development is usually related to one or more components of Virchow's triad:

✓endothelial injury

✓abnormal blood flow, stasis, or turbulence

✓hypercoagulability: either primary or secondary

- Thrombi may propagate, resolve, become organized, or embolize.
- Thrombosis causes tissue injury by local vascular occlusion or by distal embolization

THANK YOU